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Transparent Graphene Electrodes for Biomedical Applications — •Pranoti Kshirsagar[1], Thomas Chassé[3], Monika Fleischer[2], and Claus J. Burkhardt[1] — [1]NMI Natural and Medical Sciences Institute at the University of Tübingen, 72770 Reutlingen (Germany) — [2]Institute for Applied Physics and Center LISA+, Eberhard Karls University Tübingen, 72076 Tübingen (Germany) — [3]Institute of Physical and Theoretical Chemistry and Center LISA+, Eberhard Karls University Tübingen, 72076 Tübingen (Germany)

Since its isolated preparation in 2004, graphene is one of the most extensively researched materials. However, the entry of graphene into the field of biomedicine is relatively recent. Microelectrode arrays (MEAs) are often used to record the cellular activity from cells like neurons or cardiomyocytes. Au, TiN, Pt and PEDOT-CNTs are some of the electrode materials which are excellently suited for the cellular recordings. However, all these materials are opaque posing limitation for applications such as optogenetics and calcium imaging.

Here we present the development of MEAs with transparent graphene electrode and their application. 2.5 cm x 2.5 cm largely monolayer graphene is CVD grown and subsequently transferred onto the conduction lines of a MEA substrate. Detailed correlative scanning electron microscopy (SEM) and Raman spectroscopy is performed confirming the presence of graphene after each processing step. SEM is used for visualization of graphene and Raman confirms the number of layers. Graphene micro electrodes of 30 micron diameter are fabricated with this reliable process and successfully tested with cell cultures.

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